M101 · Unit 8 · Quick reference

Accumulation and the fundamental theorem
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Recall **Endpoint sums**	$\Delta x = (b-a)/n$. Left samples $a + i\Delta x$, i=0,…,n−1; right samples i=1,…,n. Multiply every height by its width.
Understand **Integral as a limit**	$\int_a^b f(x)\,dx = \lim \sum f(c_i)\Delta x_i$ as the largest width tends to zero. Continuous f on [a,b] guarantees existence.
Recall **Orientation and joining**	$\int_b^a f = -\int_a^b f$; $\int_a^a f = 0$; $\int_a^b f = \int_a^c f + \int_c^b f$.
Recall **Linearity**	$\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g$, for constants α,β.
Understand **Signs, units, bounds**	Net signed total; units = integrand units × input units. For a<b and m≤f≤M: $m(b-a) \le \int_a^b f \le M(b-a)$.
Recall **Accumulation theorem**	$A(x) = \int_a^x f(t)\,dt$ gives $A'(x) = f(x)$ at interior x when f is continuous. A(a)=0.

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Recall **Moving endpoints**	$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))g'(x)$. Continuous f, differentiable g. A moving lower limit contributes a minus sign.
Recall **Evaluation theorem**	$\int_a^b f(x)\,dx = F(b) - F(a)$, if f is continuous and F′=f on the interval.
Understand **Antiderivative family**	$\int f(x)\,dx = F(x) + C$. Equal derivatives imply a constant difference on an interval; constants cancel in endpoint subtraction.
Recall **Elementary pairs**	$x^p \mapsto \frac{x^{p+1}}{p+1}$ (p≠−1); $1/x \mapsto \ln\lvert x\rvert$; $e^x \mapsto e^x$; $\cos x \mapsto \sin x$; $\sin x \mapsto -\cos x$. Valid real intervals; reciprocal excludes zero; trig uses radians.
Recall **Initial value**	$Q(t) = Q_0 + \int_{t_0}^{t} q(s)\,ds$, for Q′=q continuous and Q(to)=Qo.
Consult **Checks**	Differentiate an antiderivative; substitute initial data; inspect the whole domain, signs, units and bounds. Benchmark code against a hand sum and an exact integral.

Consult · Python

```
def left_sum(f, a, b, n):
    # Increasing interval; positive integer n.
    width = (b - a) / n
    total = 0.0
    for i in range(n):
        total += f(a + i * width) * width
    return total

# Benchmark: f(x)=3 on [1,4] gives 9.
# Use the lesson function for guarded/oriented inputs.
```

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